Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory	
Academic Year	2025–2026 (Even)	Batch 2023–2027
Due Date	**27.02.2026**	

Experiment 2: Binary Classification using Naïve Bayes and K-Nearest Neighbors

Aim

To implement Naïve Bayes and K-Nearest Neighbors (KNN) classifiers for a binary classification problem, evaluate them using multiple performance metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

Dataset

A benchmark binary classification dataset containing numerical features and two class labels is used.

- Kaggle: Spambase Dataset

Preprocessing Steps

- The dataset was loaded into a Pandas DataFrame.
- Input features and target labels were separated.
- A stratified train–test split was performed.
- Feature scaling was applied using StandardScaler.

Implementation Details

- Implemented Gaussian, Multinomial, and Bernoulli Naïve Bayes classifiers.
- Implemented a baseline KNN classifier.
- Tuned KNN hyperparameters using GridSearchCV and RandomizedSearchCV.
- Compared KDTree and BallTree neighbor search algorithms.

Naïve Bayes Performance Comparison

Table 1: Naïve Bayes Performance Metrics

Metric	Gaussian NB	Multinomial NB	Bernoulli NB
Accuracy	0.8330	0.8004	0.8937
Precision	0.7294	0.7833	0.9096
Recall	0.9588	0.7268	0.8299
F1 Score	0.8285	0.7540	0.8679
Specificity	0.7400	0.8500	0.9400
Training Time (s)	–	–	–

Naïve Bayes Visualizations

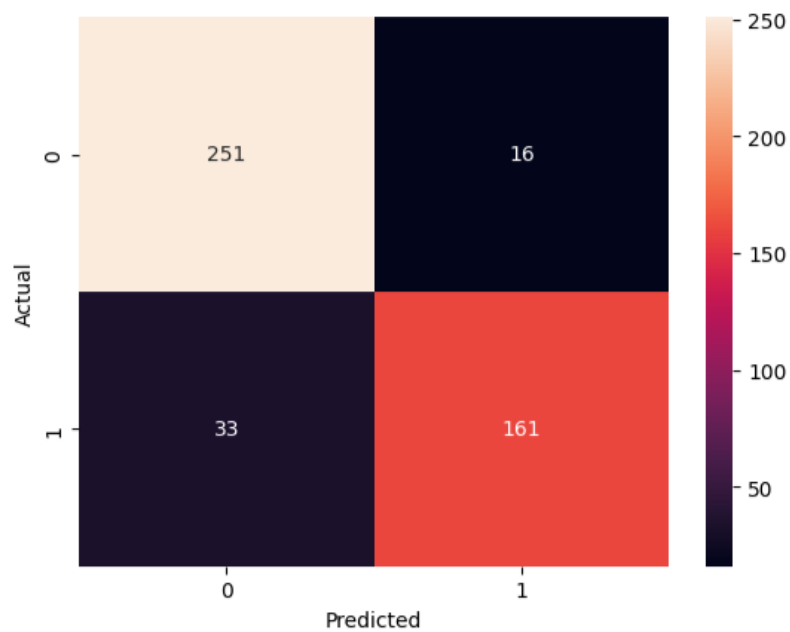

Figure 1: Confusion Matrix for Gaussian Naïve Bayes

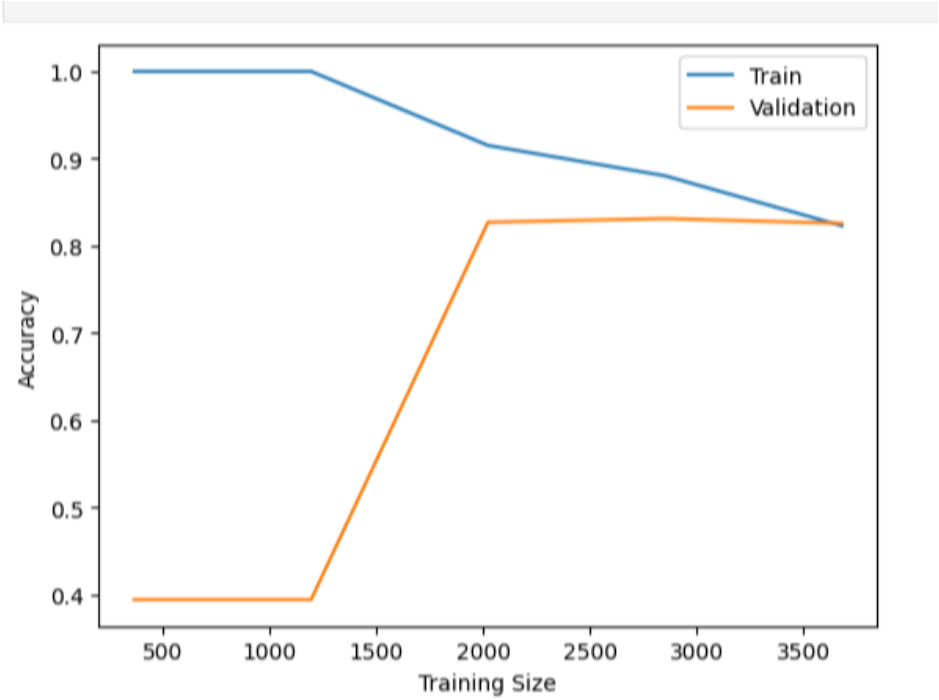

Figure 2: Training and Validation Accuracy for Gaussian Naïve Bayes

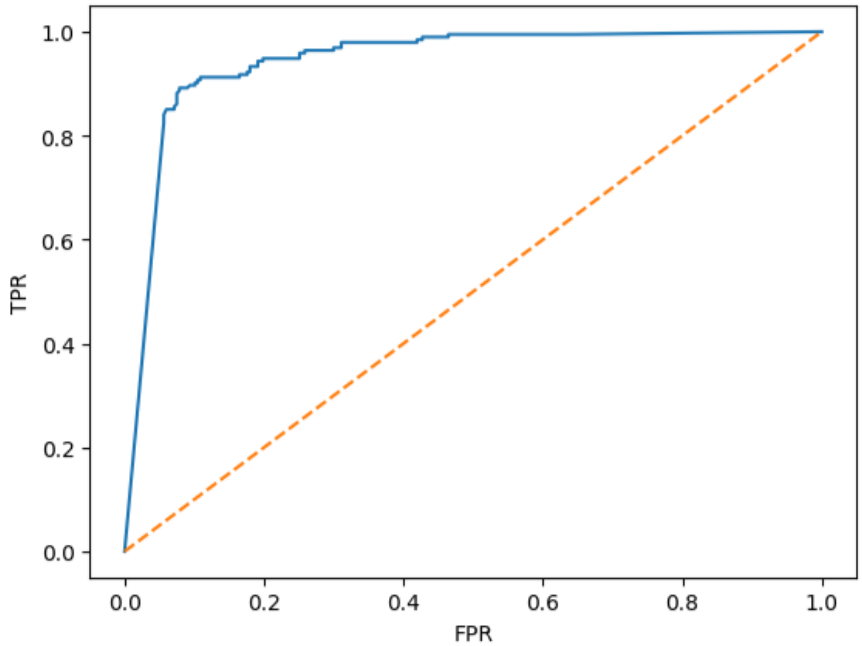

Figure 3: ROC Curve for Gaussian Naïve Bayes

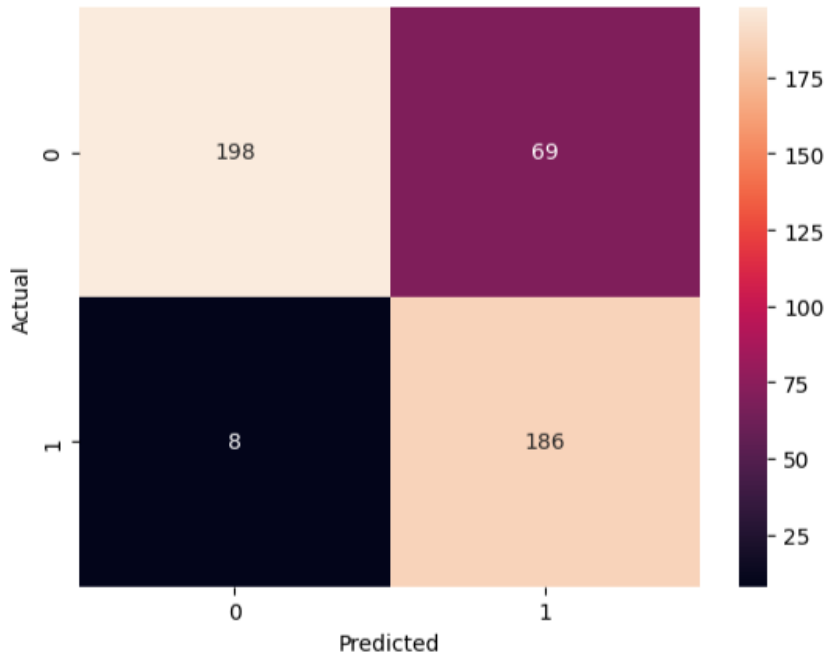

Figure 4: Confusion Matrix for Multinomial Naïve Bayes

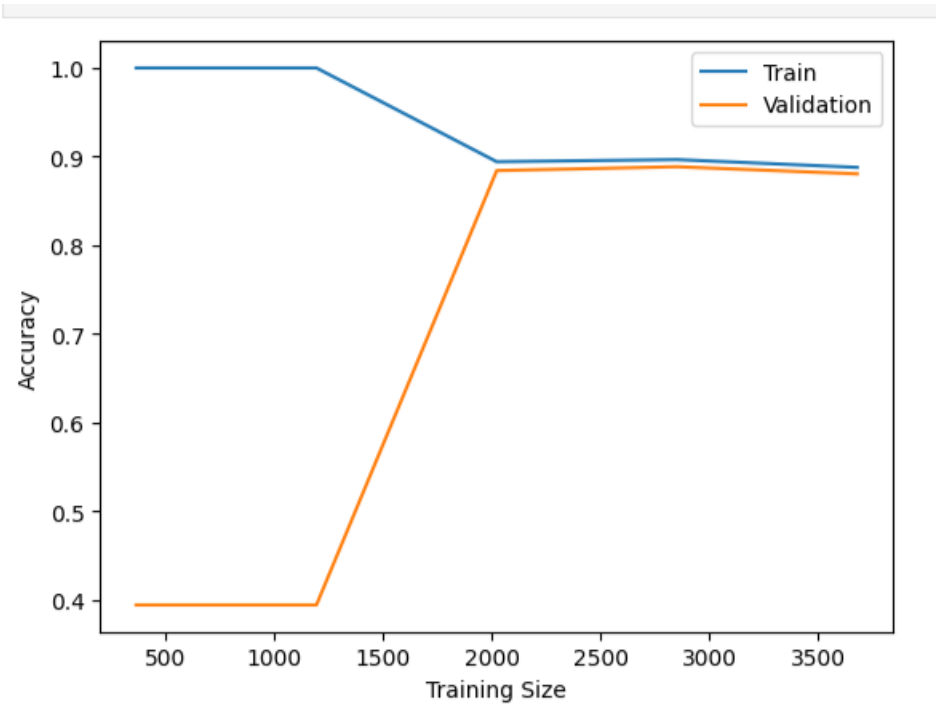

Figure 5: Training and Validation Accuracy for Multinomial Naïve Bayes

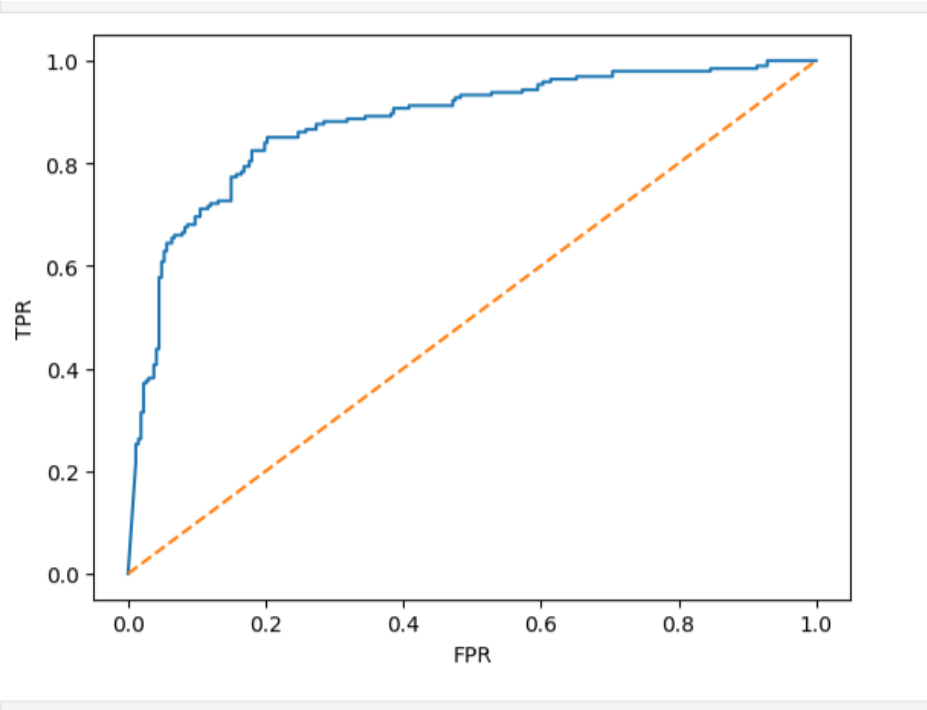

Figure 6: ROC Curve for Multinomial Naïve Bayes

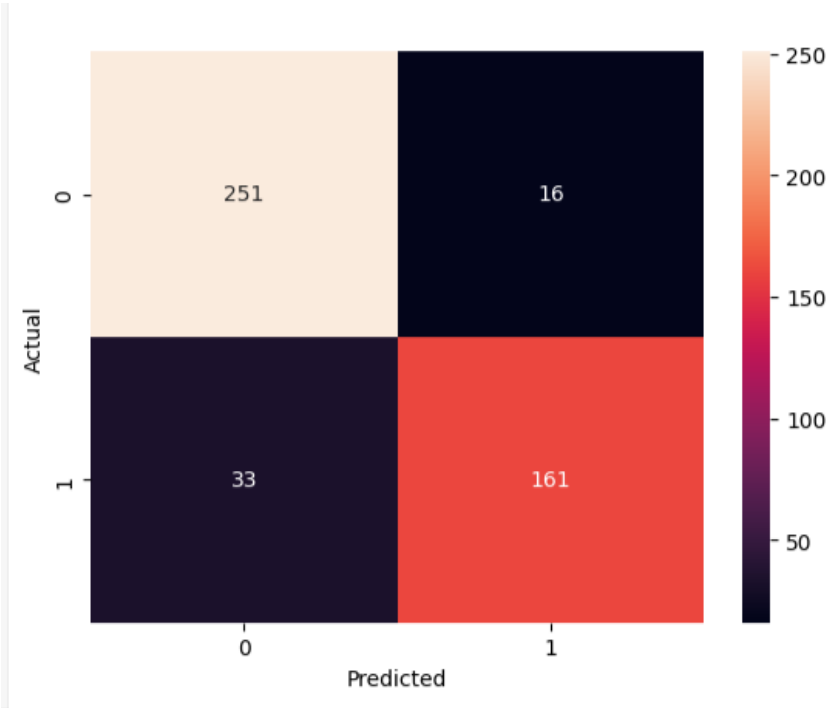

Figure 7: Confusion Matrix for Bernoulli Naïve Bayes

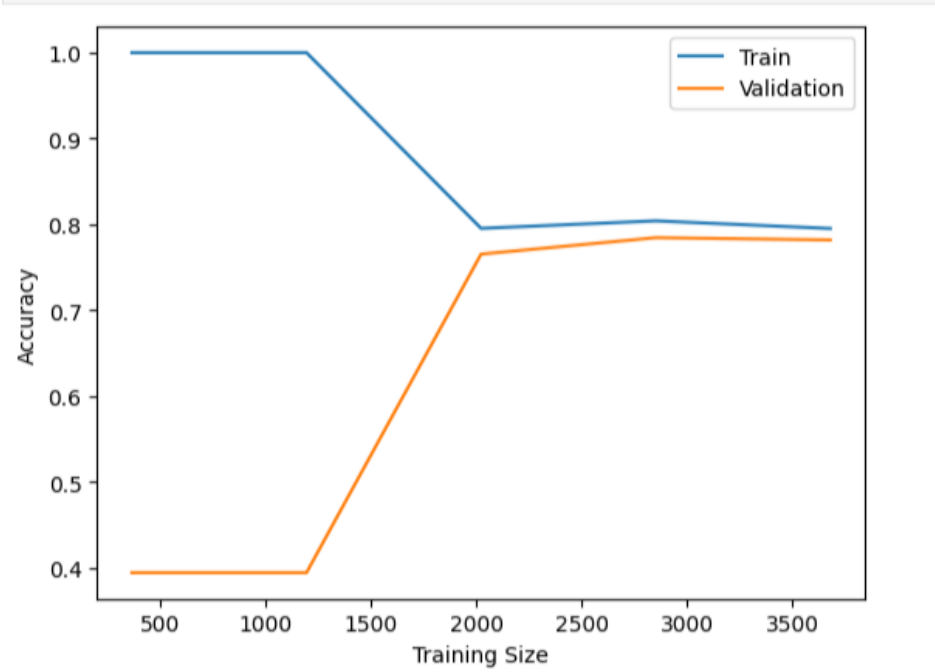

Figure 8: Training and Validation Accuracy for Bernoulli Naïve Bayes

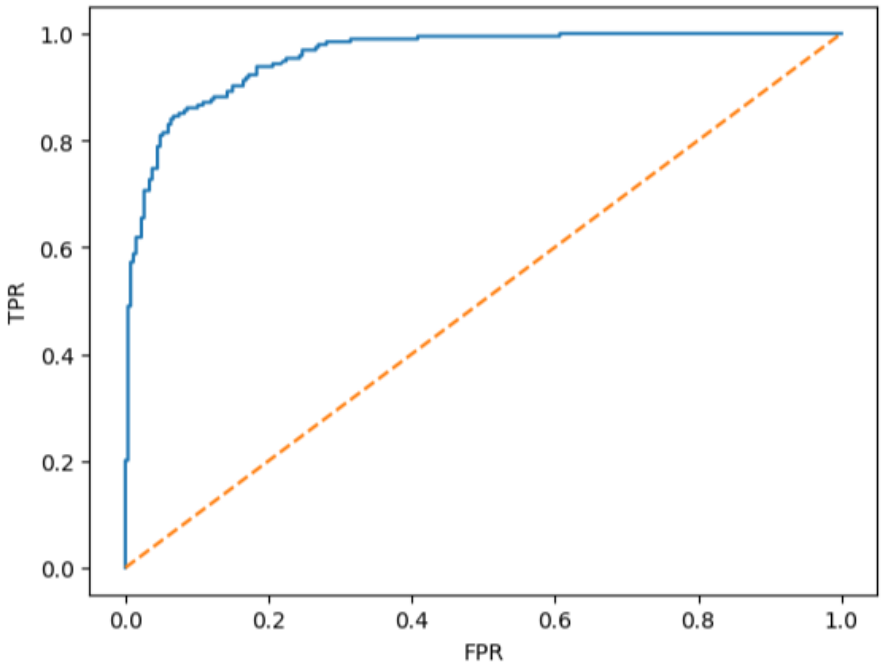

Figure 9: ROC Curve for Bernoulli Naïve Bayes

KNN Hyperparameter Tuning Results

Table 2: KNN Hyperparameter Tuning

Search Method	Best k	Best CV Accuracy	Best Parameters
Grid Search	7	0.962	{weights=distance, metric=manhattan, algorithm=kd_tree}
Randomized Search	9	0.958	{weights=distance, metric=euclidean, algorithm=ball_tree}

KNN Performance using Different Search Methods

Table 3: KNN Performance using KDTree

Metric	Value
Optimal k	7
Accuracy	0.8351
Precision	0.8349
Recall	0.8351
F1 Score	0.8340
Training Time (s)	0.02
Prediction Time (s)	0.23

Table 4: KNN Performance using BallTree

Metric	Value
Optimal k	11
Accuracy	0.8351
Precision	0.8349
Recall	0.8351
F1 Score	0.8340
Training Time (s)	0.07
Prediction Time (s)	0.016

KDTree vs BallTree Comparison

Table 5: Comparison of Neighbor Search Algorithms

Criterion	KDTree	BallTree
Accuracy	0.8351	0.8351
Training Time (s)	0.02	0.07
Prediction Time (s)	0.23	0.016
Memory Usage	Low / Medium	Medium / High

Overfitting and Underfitting Analysis

- Large gaps between training and validation accuracy indicate overfitting.
- Similar values suggest good generalization.
- Small k values cause high variance; large k values increase bias.
- Hyperparameter tuning balances the bias–variance trade-off.

Bias–Variance Analysis

- Naïve Bayes exhibits high bias and low variance due to strong assumptions.
- KNN variance depends on neighborhood size.
- Aggressive tuning reduces bias but may increase variance.

Observations and Conclusion

This experiment compared Naïve Bayes and KNN classifiers using multiple evaluation metrics and visualizations. The results highlight the impact of model complexity and hyperparameter tuning on generalization performance. While Naïve Bayes provided stable results with lower variance, KNN required careful tuning to avoid overfitting. Cross-validation proved essential for fair comparison and optimal parameter selection.